

PetVision: Cat vs Dog Image Classification

Final capstone report · Kshitij Saxena · github.com/NatKros/Image-Classification

1. Problem

Classify photos as cat or dog, and find out whether a CNN designed from scratch can match a fine-tuned pretrained network once accuracy, inference speed and model size are all taken into account.

2. Data

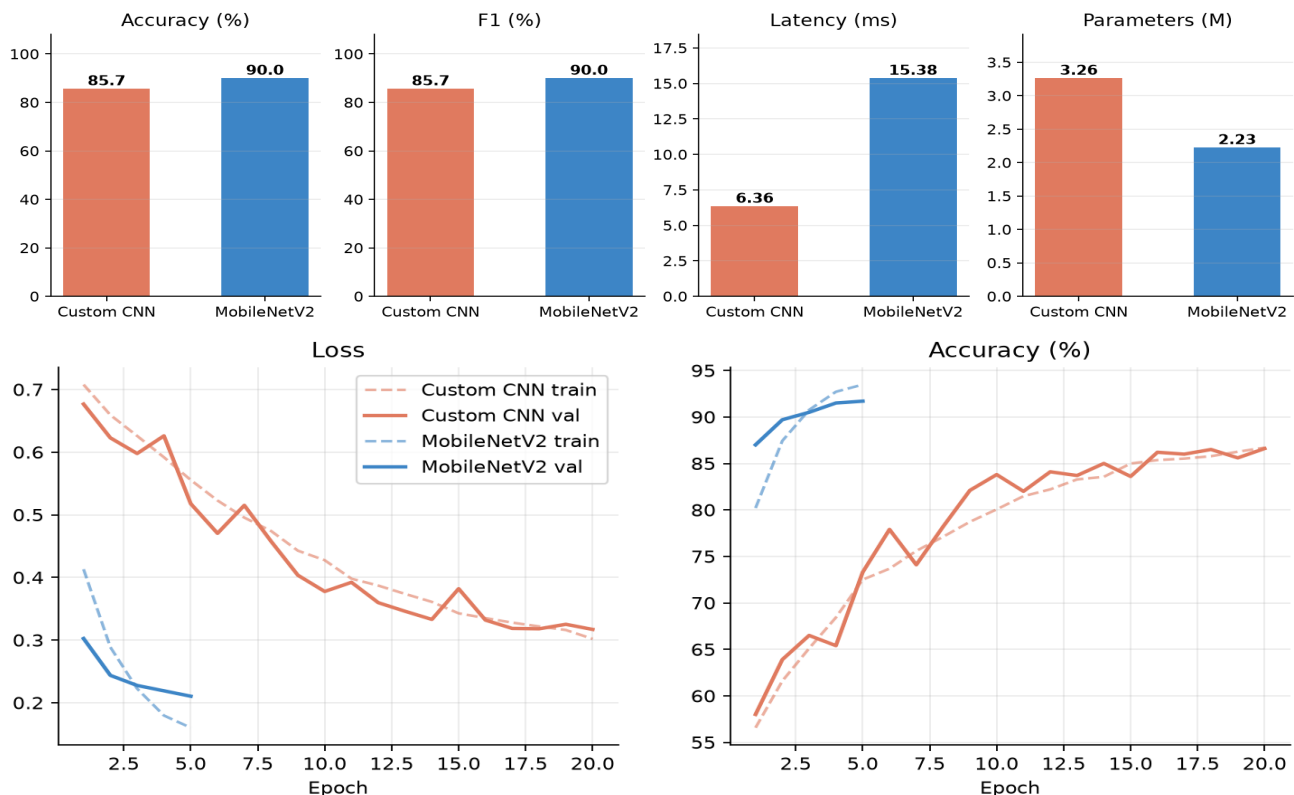
The cat and dog classes of CIFAR-10 (real photographs): 9,000 images for training, 1,000 for validation and the 2,000 official test images, which were never used for tuning. Images are upscaled to 96×96 and augmented with flips, small rotations and shifts and colour jitter.

3. Models

Custom CNN (3,263,650 parameters): four double-convolution blocks (32→256 channels) with BatchNorm, max-pooling and spatial dropout, then a small dense head. **MobileNetV2** (2,226,434 parameters): ImageNet weights with a new 2-class head, fine-tuned end to end. Both train with AdamW and cosine learning-rate decay, keeping the checkpoint with the best validation accuracy.

4. Results

Model	Accuracy	F1	Latency	Parameters	Train time
Custom CNN	85.70%	85.69%	6.4 ms	3,263,650	1912 s
MobileNetV2	90.05%	90.05%	15.4 ms	2,226,434	473 s

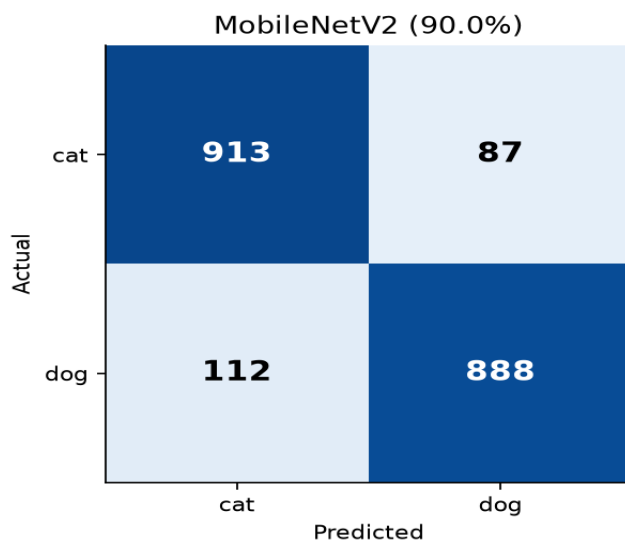
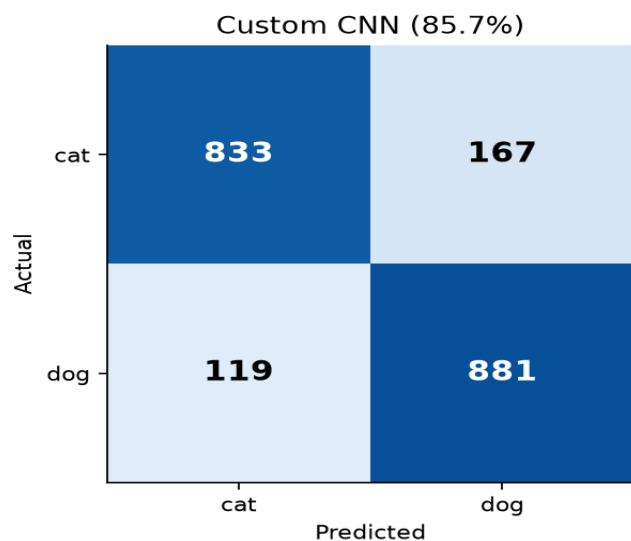
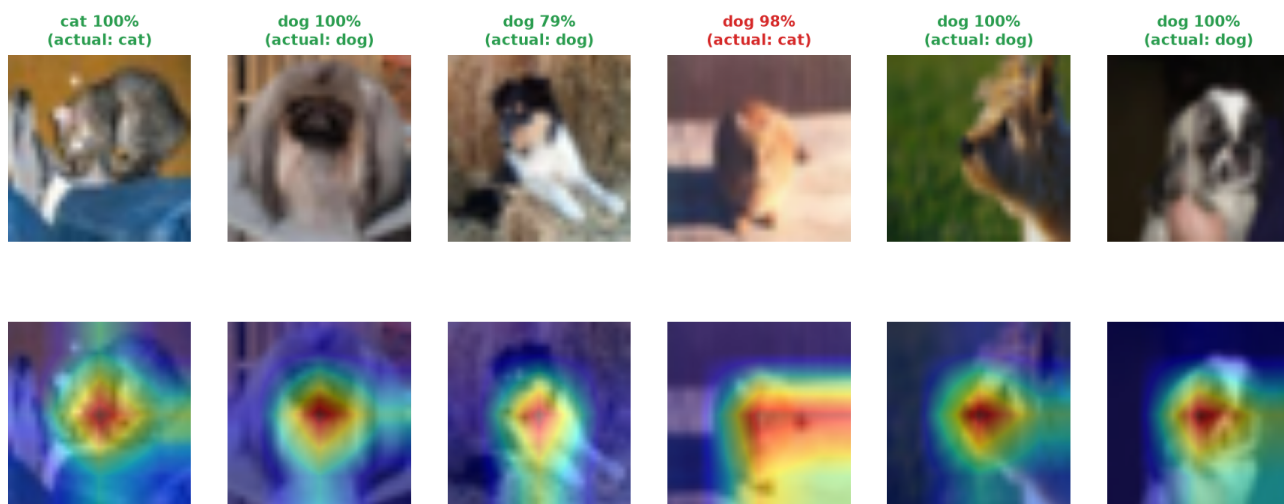


5. Interpretation

MobileNetV2 reaches 90.0% against 85.7% for the scratch CNN, trains in 473 s versus 1912 s, and classifies an image in 15.4 ms versus 6.4 ms. Pretrained features matter most when data is limited. Grad-CAM maps (below) show both networks focusing on the animal's head and body rather than the background. The remaining errors are mostly

look-alikes at 32×32 resolution.

MobileNetV2: predictions and Grad-CAM attention



6. Deployment & future work

The Streamlit app (app.py) classifies uploads with both models side by side and overlays Grad-CAM. Next steps: train on full-resolution photos (e.g. Oxford-IIIT Pets), export to ONNX, and try a Vision Transformer.